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Subject: “Also bought” recommendations and customer groups for Q4 planning

Method and Why: We turned two years of orders into one table, about 5,500 customers by 3,700 products, and compressed it with a method called SVD. SVD reduces the table to 100 buying patterns and describes every customer and product by them. Products with similar patterns become each other’s suggestions. We chose SVD over PCA because it works on our table without any modifications to it, while PCA would first have to fill in every empty cell, increasing the table size a lot. When compared, they both gave the same quality of recommendations.

Here are the customers split into four groups, based on the patterns (see customer_map.png):

- **Home & kitchen gifts** (2,000)
- **Rustic home décor** (1,400)
- **Kids & party** (1,350)
- **Bags & storage** (730)

What is lost: Compression keeps the main buying patterns and drops the fine details of individual orders, many of which are one-time purchases. To check if what was kept is enough, we hid the last two months of orders and asked the system to predict them. For 38% of returning customers, a product they bought for the first time in that period was in their top 10 suggested items. Suggesting our 10 bestsellers instead worked for 27%.

Size and speed: The full table would take 161 MB, the compressed version takes 7MB, 22 times smaller. The whole nightly job, which rebuilds the patterns, the product suggestions and a top-10 list for every customer, runs in about one second.

Risks:

- New customers and new products have no history, therefore they get no suggestions until they accumulate enough data.
- About a fifth of order data entries have no customer ID and couldn’t be used.
- We tested during busy gift season, so expectations for the quiet months could be different.
- Suggestions lean towards popular products, so lesser-known items appear less.

Next steps:

- Trial the box on part of the site and compare sales against best sellers.
- Show best sellers to new customers until they have a purchase history.
- Re-check the customer groups each quarter.